Caleb Kang

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Education

University of Illinois Urbana-Champaign — Urbana, IL
Bachelor of Science in Computer Science, Grainger College of Engineering — *Aug. 2026 – May 2028*

- *In Progress (Fall 2026):* **Computer Architecture** (CS 233), **Database Systems** (CS 411), Software Design Lab (CS 222), Probability & Statistics for Computer Science (CS 361)

William Rainey Harper College — Palatine, IL
*Associate in Engineering Science, GPA: **3.88/4.0*** — *Aug. 2024 – May 2026*

- *Completed Coursework:* Data Structures & Algorithms, Linear Algebra, Differential Equations, Discrete Mathematics
- *Awards:* **Motorola Solutions Foundation Scholarship (Full-Ride)**, Engineering Pathways Scholarship, President's List

Experience

Software Development Engineer Intern — May 2026 – Aug. 2026
Amazon — *Bellevue, WA*

- Built an **LLM** investigation agent for Teller, Amazon's Accounts Payable system (**$2.5B+** processed), turning analyst questions into validated **SQL** over **130M+** production records
- Hardened the tool layer beneath the LLM with a read-only **SQL** validator, table allowlist, **Bedrock guardrails**, and anti-hallucination grounding checks; shipped **100+** unit and integration tests
- Automated anomaly detection with **z-score** baselines and a **20+** tool-call investigation loop; surfaced a production void batch of **63,469** payments (z-score **62**) in 2–3 minutes
- Deployed the containerized agent on **Bedrock AgentCore** with **AWS CDK** (TypeScript) infrastructure as code, **CI/CD**, and least-privilege cross-account **IAM** access to SOX-regulated data

Community College Intern — Jun. 2025 – Aug. 2025
Fermi National Accelerator Laboratory — *Batavia, IL*

- Built a fault-tolerant **Python** state machine to orchestrate **ASIC** test workflows for the Deep Underground Neutrino Experiment (DUNE), supporting validation of approximately **300,000** chips across **six** national labs
- Mapped state transitions and failure paths in **UML**, implemented error handling, and defended the architecture in peer reviews and technical presentations to research scientists

Student Researcher — Oct. 2024 – May 2025
Biola University (Remote) — *La Mirada, CA*

- Fine-tuned and benchmarked a multimodal authentication pipeline (**CLIP**, **Whisper**, **Sortformer**) in **PyTorch** on **NVIDIA NeMo**
- Built evaluation workflows measuring **inference latency**, **throughput**, and edge-case reliability across vision, audio, and speaker-diarization inputs

Projects

C++ Farming Simulator | *C++, OOP, Memory Management, Catch2* — Oct. 2025 – Dec. 2025

- Built a terminal-based simulation on a dynamic **2D pointer grid**, handling **heap allocation and deallocation** manually across gameplay state changes
- Covered state invariants and edge cases in movement and resource logic with a deterministic **Catch2** test suite

Duckiebot Robotics Initiative | *Linux, Docker, C++, Raspberry Pi, NVIDIA Jetson* — Dec. 2024 – Jan. 2025

- Led a **4-person** team building autonomous rover controls in **C++**; owned sprint planning and deployment validation checks
- Deployed the Duckietown stack in **Docker** on embedded **Linux** (Raspberry Pi, **NVIDIA Jetson**), debugging kernel-level networking, serial communication, and sockets

Technical Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript
Systems & Infrastructure: Linux/Unix, Docker, Git, CI/CD, AWS (Athena, Glue, Lake Formation, S3, Bedrock, CDK)
Data & ML: Relational Databases, PyTorch, NVIDIA NeMo, Strands Agents SDK
Concepts: Data Structures & Algorithms, OOP, State Machines, Memory Management, Containerization, Unit Testing